

Rocket Mission Simulator — Product Backlog

#	Name	How to Demo	Notes
1	Display rocket simulation	Launch the application; the rocket and its environment are displayed in the simulation window.	Display a 2D rocket and its surroundings. The environment should visually change as the rocket travels from the atmosphere into space.
2	Launch the rocket	Start a new mission and launch the rocket; the rocket begins travelling upward and the displayed flight information updates.	Initialise the rocket and mission state when starting a new mission.
3	Control the rocket	During flight, adjust the rocket's throttle and direction; its movement changes in response to the controls.	Allow the user to control throttle and rocket orientation. Controls should influence the simulated flight. The rocket orientation should be changed through two buttons (left and right tilt buttons) and the throttle through a throttle slider that goes from 0 to 100%.
4	Simulate rocket flight	Launch and control the rocket; its position and velocity change throughout the flight according to its current state and user input.	Use a simplified flight model considering factors such as <u>thrust, gravity, fuel and atmospheric effects</u> . Realistic rocket physics are not required for the purpose of the MVP.
5	Display flight telemetry	Fly the rocket; information about its current flight is displayed and continuously updated (for which we will use Multithreading)	Display relevant information such as altitude, velocity, fuel, throttle and orientation.
6	Follow the rocket during flight	Launch the rocket and continue ascending; the simulation view follows the rocket and adapts as its altitude increases.	Keep the rocket visible throughout the flight. The view may change its scale as the rocket travels farther from Earth (zooming out the frame).
7	Handle mission events	During the mission, encounter an event; the application displays the event and allows the user to respond to it.	Events can include situations such as <u>overheating, excessive aerodynamic stress, guidance problems or system anomalies</u> . Some events may require the user to adjust the flight controls. *Details about event are to be determined later (we got to do some rocket research).
8	Recover from mission events	Trigger a recoverable event and perform the appropriate action; the warning disappears and the mission can continue.	Player actions should affect event outcomes. Ignoring serious problems may negatively affect the rocket or cause mission failure. *To be determined later what actions/states will cause mission failure (have to do rocket research)
9	Reach the target orbit	Fly the rocket to the required conditions; the application detects that the target orbit has been reached.	Use simplified conditions based on factors such as altitude and velocity rather than full orbital mechanics.
10	Deploy the satellite	Reach the target orbit and deploy the satellite; the satellite separates from the rocket and is displayed in orbit.	Satellite deployment is the main objective of the mission. Its movement in orbit can be represented visually rather than through realistic orbital physics.
11	Display mission outcome	Successfully deploy the satellite or cause the mission to fail; the application displays the appropriate result.	Handle both successful and unsuccessful mission outcomes and prevent the mission from continuing after it has ended.
12	Display mission log	Perform several actions and encounter events during a mission; the application displays a record of important events.	Record relevant information such as launch, warnings, event outcomes, orbit achievement and satellite deployment.
13	Restart the mission	Complete or fail a mission and select Restart; the simulator returns to its initial state and a new mission can begin.	Reset the rocket, telemetry, events, mission state and other relevant application state.